DRAWING LIST

ITEM	QTY	DRAWING NUMBER	DESCRIPTION
1	1	S1101EFA3500	AVIONICS POWER HARNESS ASSEMBLY
2	1	S1101EFA3501	AVIONICS BENCH CONNECTION/PIL1 HARNESS ASSEMBLY
3	1	S1101EFA3502	CARD STACK HARNESS ASSEMBLY
4	1	S1101EFA3503	HARD DRIVE POWER CABLE ASSEMBLY
5	1	S1101EFA3504	EXTERNAL INTERFACE VGA HARNESS ASSEMBLY
6	1	S1101EFA3505	DIGITAL CAMERA INTERFACE HARNESS ASSEMBLY
7	1	S1101EFA3506	HARD DRIVE SATA DATA CABLE ASSEMBLY
8	1	S1101EFA3507	EXTERNAL INTERFACE USB HARNESS ASSEMBLY
9	1	S1101EFA3508	RS232 MOTION CONTROL BOARD HARNESS ASSEMBLY
10	1	S1101MFA3100	AVP HOUSING ASSEMBLY
11	1	S1101MFA3119	HOUSING REAR FRAME INTERFACE BRACKET ASSEMBLY
12	1	S1101MFA3127	HOUSING TOP COVER ASSEMBLY
13	1	S1101MFA3131	BREAKER COVER ASSEMBLY
14	1	S1101MFA3133	CONNECTOR COVER ASSEMBLY
15	1	S1101MFA3200	CARD CAGE ASSEMBLY
16	1	S1101MFA3300	HARD DRIVE ASSEMBLY
17	1	S1101MFA3305	HARD DRIVE TRAY ASSEMBLY
18	1	S1101MFA3400	POWER DISTRIBUTION ASSEMBLY
19	1	S1101MFD3106	AIR INLET VENT SCREEN FRAME
20	1	S1101MFD3107	AIR INLET VENT SCREEN
21	4	S1101MFD3109	SPACER WASHER
22	1	S1101MFD3121	HOUSING REAR INTERFACE PANEL
23	1	S1101MFD3124	HOUSING LEFT COVER
24	1	S1101MFD3125	HOUSING RIGHT COVER
25	1	S1101MFD3126	HOUSING FRONT COVER
26	4	S1101MFP3108	MODIFIED SOCKET HEAD CAP SCREW 1/4-20 X 1.00
27	1	S1101MFD3105	HEATSINK AIRFLOW DIVERTER
28	1	S1101EFA3216	EXTERNAL BATTERY ASSEMBLY
29	1	S1101MFT3000-XX	COMBO LABEL
30	2	S1101MFT3000-04	LOCATION LABEL "C"
31	2	S1101MFT3000-05	LOCATION LABEL "D"
32	1	S1101MFT3000-06	CONNECTOR ID LABEL
33	1	S1101MFT3000-07	CIRCUIT BREAKER LABEL
34	1	S1101MFT3000-08	CIRCUIT BREAKER LABEL
35	1	S1101MFT3000-09	CIRCUIT BREAKER LABEL

COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION
C1	3	MS3320-7-1/2	CIRCUIT BREAKER, AIRCRAFT, TRIP-FREE PUSH PULL, 7.5 AMP, TYPE1.
C2	4	JAX103-M09-F09-440	JACKSCREW, #4-40 X .312 LG THD. 303 STAINLESS, PASSIVATED, UNICORP
C3	1	809-199-8	MOUNTING FLANGE, 3/4 PERIMETER 800 SERIES CONNECTORS, GLENAIR
C4	4	NAS1291C04M	NUT, .112-40 UNJC-3B, SELF LOCKING, A286
C5	2	ECF504-UAAS	PANEL MOUNTED USB COUPLER, SHIELDED TYPE A/TYPE A, L-COM
C6	1	CSD-2	NUTPLATE, D-SUB, SHELL 15, STROCO
C7	1	DELETED	DELETED
C8	4	NAS1101E02-5	SCREW, MACHINE-FLAT, FILLISTER HEAD, CRES, PASSIVATED .0860-56 UNJC-3A X .312 LENGTH
C9	4	NAS1102E02-4	.086-56 UNJC-3A X .250", 100° FLAT HEAD, OFFSET CRUCIFORM MACHINE SCREW
C10	10	NAS1102E04-6	.112-40 UNJC-3A X .375", 100° FLAT HEAD, OFFSET CRUCIFORM MACHINE SCREW
C11	5	NAS1102E04-8	.112-40 UNJC-3A X .500", 100° FLAT HEAD, OFFSET CRUCIFORM MACHINE SCREW
C12	2	NAS1352N04-3	SCREW, CAP, SOCKET HEAD, A286, PASSIVATED .1120-40 UNRC-3A X .188
C13	44	NAS1352N04-6	SCREW, CAP, SOCKET HEAD, A286, PASSIVATED .1120-40 UNRC-3A X .375
C14	18	NAS1352N04-8	SCREW, CAP, SOCKET HEAD, A286, PASSIVATED .1120-40 UNRC-3A X .500
C15	4	NAS1352N06-8	SCREW, CAP, SOCKET HEAD, A286, PASSIVATED .1380-32 UNRC-3A X .500
C16	12	NAS1352N06-10	SCREW, CAP, SOCKET HEAD, A286, PASSIVATED .1380-32 UNRC-3A X .625
C17	2	NAS620C4	#04 FLAT WASHER REDUCED OUTER DIAMETER
C18	68	NAS620C4L	#04 REDUCED THICKNESS FLAT WASHER REDUCED O.D.
C19	16	NAS620C6L	#06 REDUCED THICKNESS FLAT WASHER REDUCED O.D.
C20	3	NAS1149C0716R	CRES WASHER, PASSIVATE FINISH .437 NOMINAL DIAMETER .016 THICK
C21	A/R	RZC02-015	GASKET, .25 X .25 X .065 W, VITON, 75 DURO, ZATKOFF
C22	A/R	MS-907	EPOXY ADHESIVE, MILLER STEPHENSON
C23	A/R	3145 GRAY	RTV ADHESIVE/SEALANT, DOW CORNING
C24	A/R	2216 B/A	2216 B/A GRAY EPOXY 3M SCOTCH WELD EPOXY
C25	A/R	TC105	CABLE TIE MOUNTING BASE, ALUMINUM, THOMAS & BETTS
C26	A/R	PLT1M-M702	HALAR TIE WRAP, 4 IN. LONG PANDUIT CORPORATION
C27	A/R	242	THREADLOCKER, LOCTITE
C28	A/R	222MS	THREADLOCKER, LOCTITE
C29	1	DELETED	DELETED
C30	4	NAS620C2	#02 FLAT WASHER REDUCED OUTER DIAMETER
C31	1	SDG32104805-009	DECAL, ELECTROSTATIC SYMBOL .5 IN X .5 IN, METAL PHOTO AND ADHESIVE (JSC)
C32	1	667-262-M-N10	CONNECTOR CAP, GLENAIR
C33	1	SDG32104806-008	DECAL, ELECTROSTATIC DISCHARGE IVA 2.88 IN X 1.25 IN, METAL PHOTO AND ADHESIVE (JSC)

15. REFER TO THE DRAWINGS FOR PC BOARD PINOUT REFERENCES:
    - S1101EFA3208 MODIFIED SINGLE BOARD COMPUTER
    - S1101EFA3209 MODIFIED ETHERNET BOARD
    - S1101EFA3210 MODIFIED POWER BOARD
    - S1101EFA3211 MODIFIED SERIAL BOARD
    - S1101EFA3212 MODIFIED MOTION CONTROL BOARD
    - S1101EFA3213 MODIFIED VGA BOARD
    - S1101EFA3214 VGA INTERFACE BOARD ASSY
    - S1101EFA3215 MOTION CONTROL BOARD INTERFACE ASSY
    - S1101EFA3310 HARD DRIVE TRAY BOARD ASSY
    - S1101EFA3403 POWER DISTRIBUTION BOARD ASSY
    - S1101EFA3404 EMI FILTER BOARD ASSY
16. CLEAN ALL SURFACES WITH ISOPROPYL ALCOHOL AND LET AIR DRY. WIPE WITH CLEAN COTTON SWABS AND/OR LINT-FREE CLOTH.
17. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
18. BEFORE STORAGE, PROTECT WITH CONTAMINATION BARRIER. USE CLEAN, NEW, ESD BAG, TOTES OR APPROVED ESD PACKAGING. MARK PART NUMBER, A DASH, AND A SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/SOW, BY TAGGING PER SAE-AS478-35D.
19. DISCARD NUTS & WASHERS SUPPLIED WITH ITEM C2.
20. APPLY LABEL TO CLEAN, DRY SURFACE IN LOCATION SHOWN.
21. SEE SHEETS 2 THROUGH 6 FOR ADDITIONAL INFORMATION.

NOTES:

1. CERTIFICATE OF CONFORMANCE REQUIRED FOR ALL COMMERCIAL PARTS.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. STAKING, BONDING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.1
4. HARNESS ASSEMBLY AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.4.
5. ASSEMBLE AND TEST PER ACME-TST-009.
6. REFER TO THE FOLLOWING DRAWINGS FOR WIRING INTERCONNECTION:
    - S1101EFA3500 AVIONICS POWER HARNESS ASSY
    - S1101EFA3501 AVIONICS BENCH CONNECTION/PIL1 HARNESS ASSY
    - S1101EFA3502 CARD STACK HARNESS ASSY
    - S1101EFA3503 HARD DRIVE POWER CABLE ASSY
    - S1101EFA3504 EXTERNAL INTERFACE VGA HARNESS ASSY
    - S1101EFA3505 DIGITAL CAMERA INTERFACE HARNESS ASSY
    - S1101EFA3506 HARD DRIVE SATA DATA CABLE ASSY
    - S1101EFA3507 EXTERNAL INTERFACE USB HARNESS ASSY
    - S1101EFA3508 RS232 MOTION CONTROL BOARD HARNESS ASSY
    - S1101EFC3509 AVIONICS INTERCONNECT DIAGRAM
7. CHEMICAL CONVERSION COAT ANY DAMAGED OR MACHINED AREAS ON ALUMINUM PARTS PER MIL-DTL-5541, TYPE I, CLASS 3.
8. APPLY THREAD LOCKING COMPOUND PER ESTABLISHED PROCEDURES.
9. TORQUE VALUES NOTED ON THE DRAWING ARE EFFECTIVE TORQUE ONLY. RUNNING TORQUE IS REQUIRED FOR SELF LOCKING DEVICES AND BOLT FRICTION. RUNNING TORQUE MUST BE MEASURED AND ADDED TO THE EFFECTIVE TORQUE. RECORD RUNNING TORQUE AND FINAL TORQUE ON DRAWING.
10. CABLE AND WIRE ROUTING AND CABLE TIE MOUNT PLACEMENT TO BE DETERMINED BY TECHNICIAN AT TIME OF ASSEMBLY.
11. REMOVE ALL LOCKWASHERS FROM BREAKER ASSEMBLY AND DISCARD. REPLACE FRONT MOUNTING LOCKWASHER WITH ITEM# C20.
12. CUT INTERFACE GASKET MATERIAL, ITEM C21, TO LENGTH OF 24.44 IN, MEASURED FROM ITS CENTER. SPLICE TOGETHER AND BOND INTO MACHINED GROOVE ON BOTTOM SIDE OF AVP HOUSING ASSEMBLY PER MANUFACTURERS INSTRUCTIONS, USING ITEM C22.
13. ALL HARNESS CONNECTIONS TO BE STAKED USING RTV (ITEM C23) AT FINAL ASSEMBLY.
14. COVER (ITEM 14) TO BE REMOVED BEFORE INSTALLATION ON OPTICS BENCH.

ACME-ECO-682 In Process

REV	DESCRIPTION	APPROVED
C	INCORPORATED ACME-ECO-633.	LF 12/06/16 AK 12/07/16 AD 12/07/16
B	INCORPORATED ACME-ECO-474, ACME-ECO-502 AND ACME-ECO-511	AD 5/10/16 JO 5/12/16 AR 5/13/16
A	INCORPORATED ACME-ECO-179	KR 10/3/14 AD 10/6/14 GF 10/9/14

REVISION HISTORY

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± .03
   .XX ± .01
   .XXX ± .005
5. ANGULAR TOLERANCE ± .5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: E.McCLOSKEY 12/08/16
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	K.RHODES 4/25/13	QUALITY	D.SEELEY 4/26/13
REVIEWER	M.PREVITE 4/26/13	MATERIALS	M.WILKINSON 4/25/13
MECHANICAL	A.DRAKE 4/25/13	PROJ LEAD	B.BOROWSKI 4/29/13
ELECTRICAL	T.GOBELI 4/25/13		
STRUCTURAL	C.WERNER 4/25/13		
SAFETY	W.BORRELLI 4/26/13		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

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CONTRACT NUMBER: NNC09BA02B
PROJECT CODE: SPACE110

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE: ACME AVIONICS PACKAGE

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA3000	C

SCALE 1/2 | INITIAL REL DATE 4/29/13 | SHEET 1 OF 6

RELEASE LEVEL: RELEASED
MODEL: S1101MFA3000

PRO/E VERSION: Wildfire 2.0

REV C | SH 1 | DWG NO S1101MFA3000

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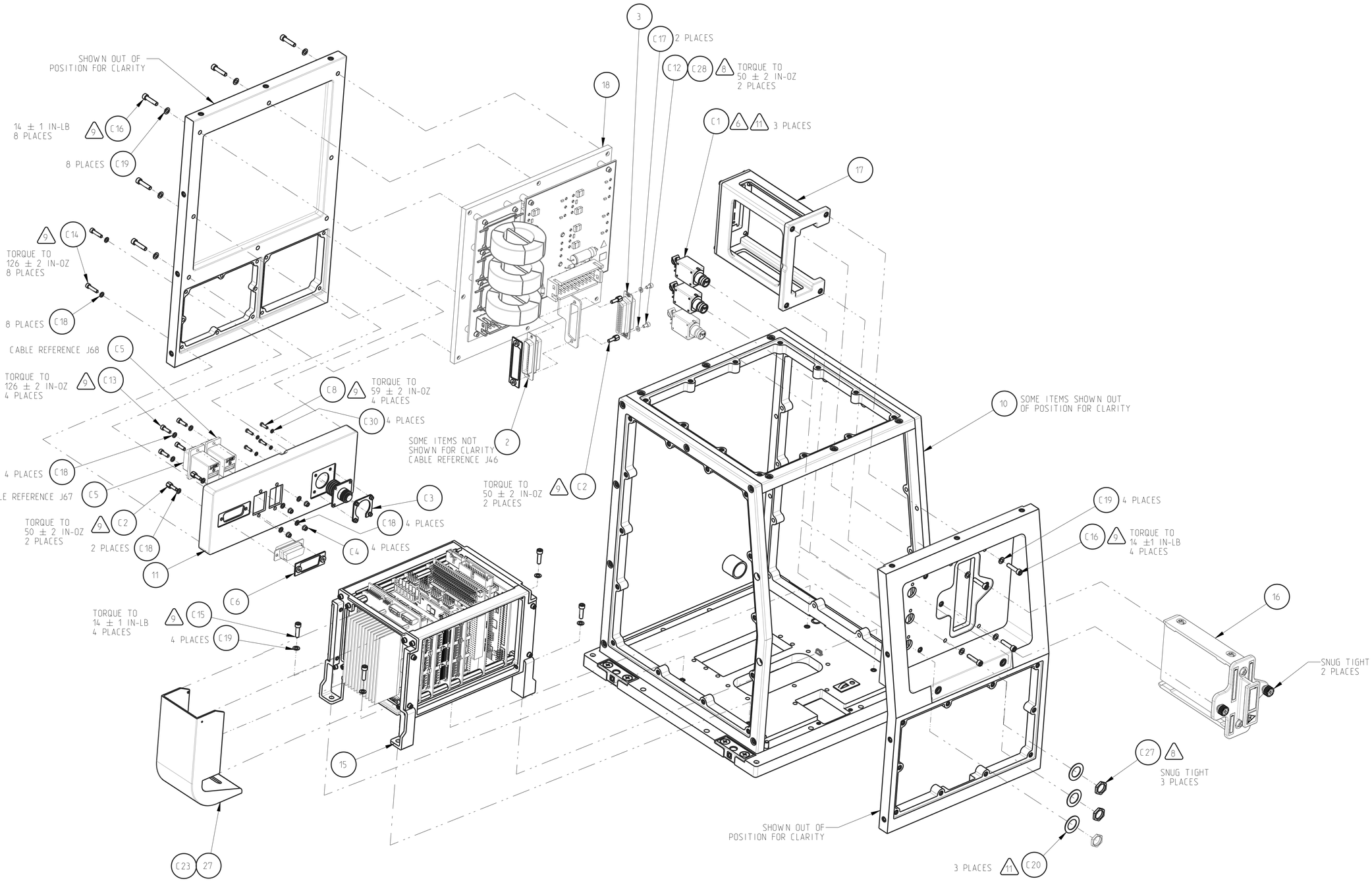
NOTES:

1. SEE SHEET 1 FOR ADDITIONAL INFORMATION. GENERAL NOTES AND TOLERANCES.



COMPONENT ASSSEMBLY VIEW

ACME-ECO-682 In Process

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA3000	C
SCALE 1/2	INITIAL REL DATE 4/29/13	SHEET 2 OF 6	

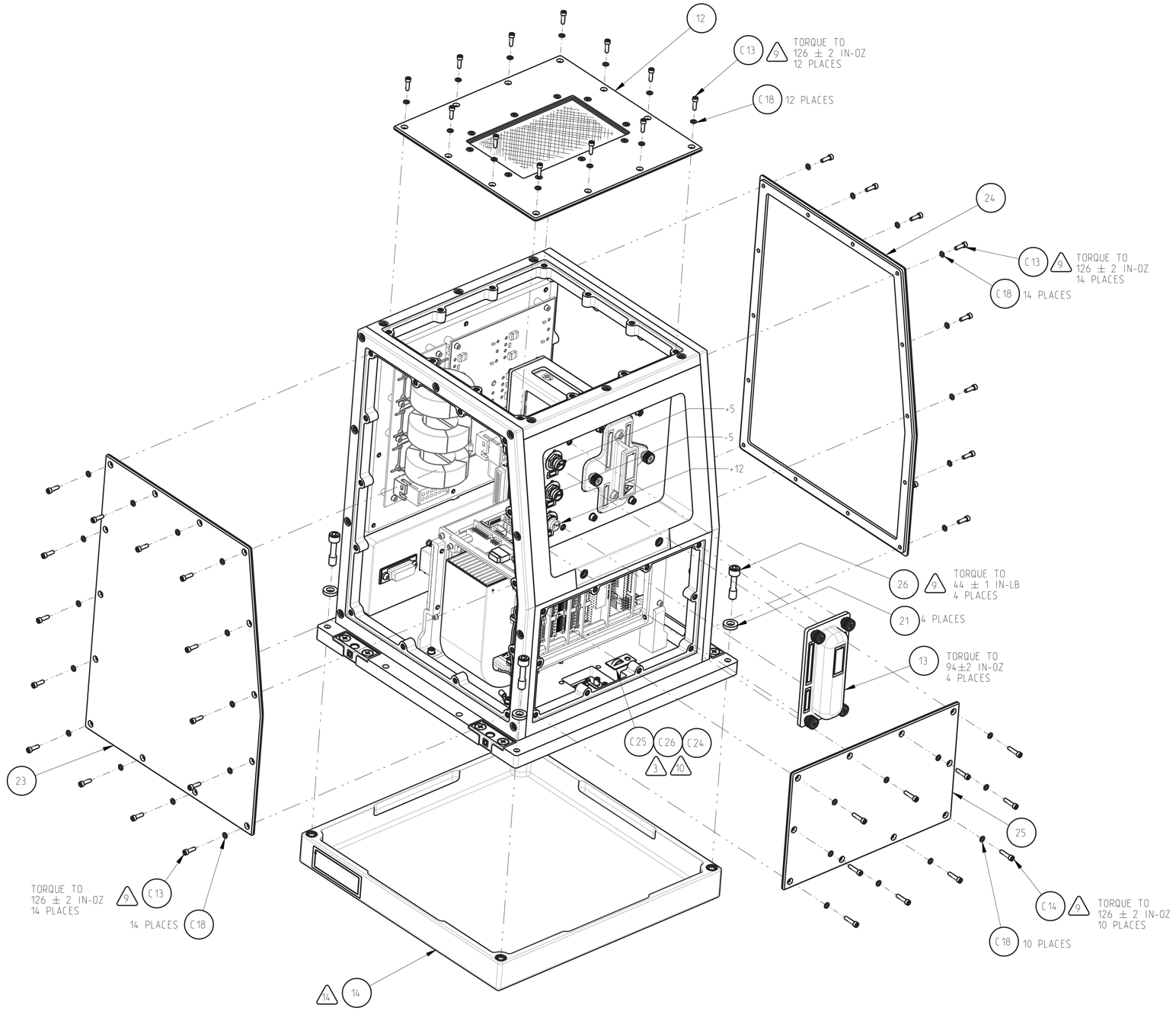
PRO/E VERSION: Wildfire 2.0

ZIN_D_2012.frm

DWG NO S1101MFA3000 REV C SH 2

NOTES:

1. SEE SHEET 1 FOR ADDITIONAL INFORMATION. GENERAL NOTES AND TOLERANCES.



PANEL ASSEMBLY VIEW

ACME-ECO-682 In Process

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA3000	C
SCALE 1/2	INITIAL REL DATE 4/29/13	SHEET 3 OF 6	

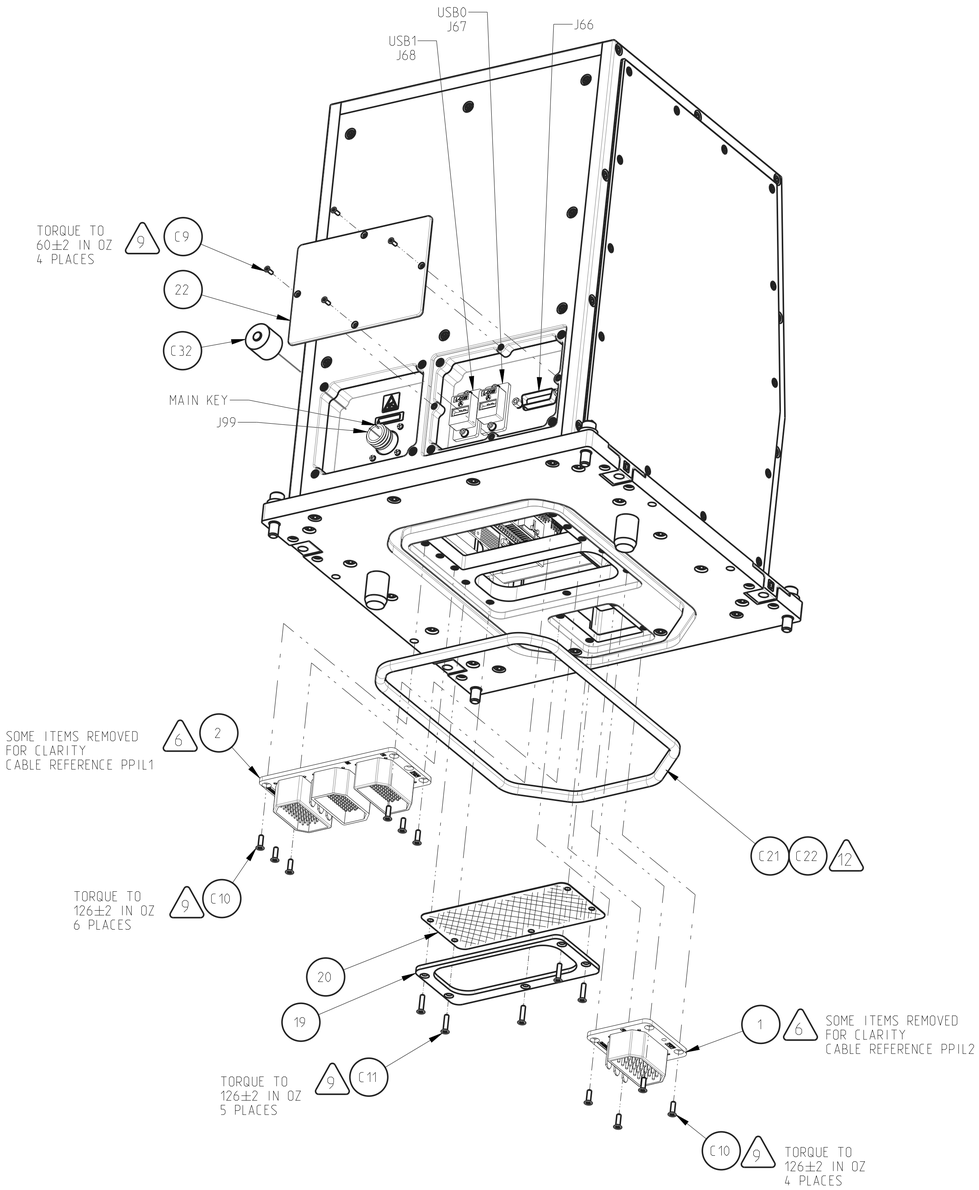
PRO/E VERSION: Wildfire 2.0

DWG NO S1101MFA3000 | SH 3 | REV C

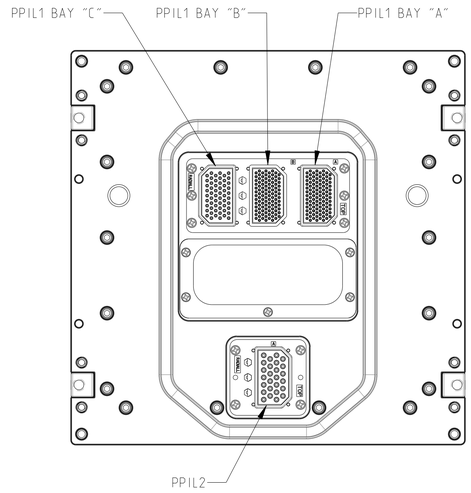
ZIN_D_2012.frm

NOTES:

1. SEE SHEET 1 FOR ADDITIONAL INFORMATION. GENERAL NOTES AND TOLERANCES.



INTERFACE COMPONENTS VIEW



CONNECTOR ORIENTATION VIEW

ACME-ECO-682 In Process

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA3000	C
SCALE 1/2	INITIAL REL DATE 4/29/13	SHEET 4 OF 6	

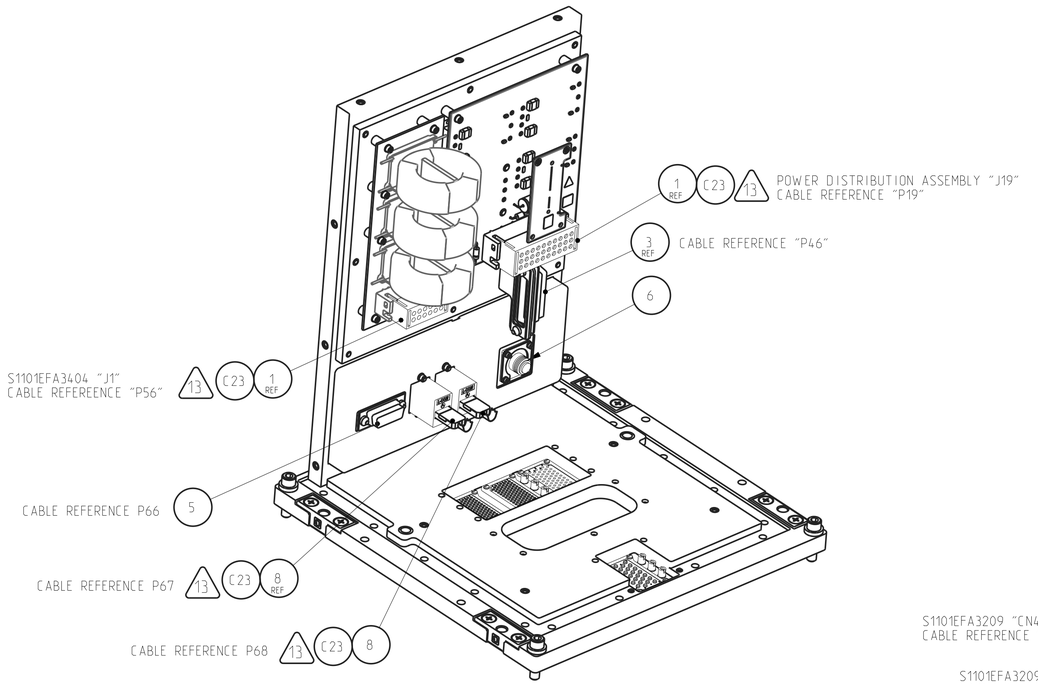
PRO/E VERSION: Wildfire 2.0

ZIN_D_2012.frm

DWG NO S1101MFA3000 | SH 4 | REV C

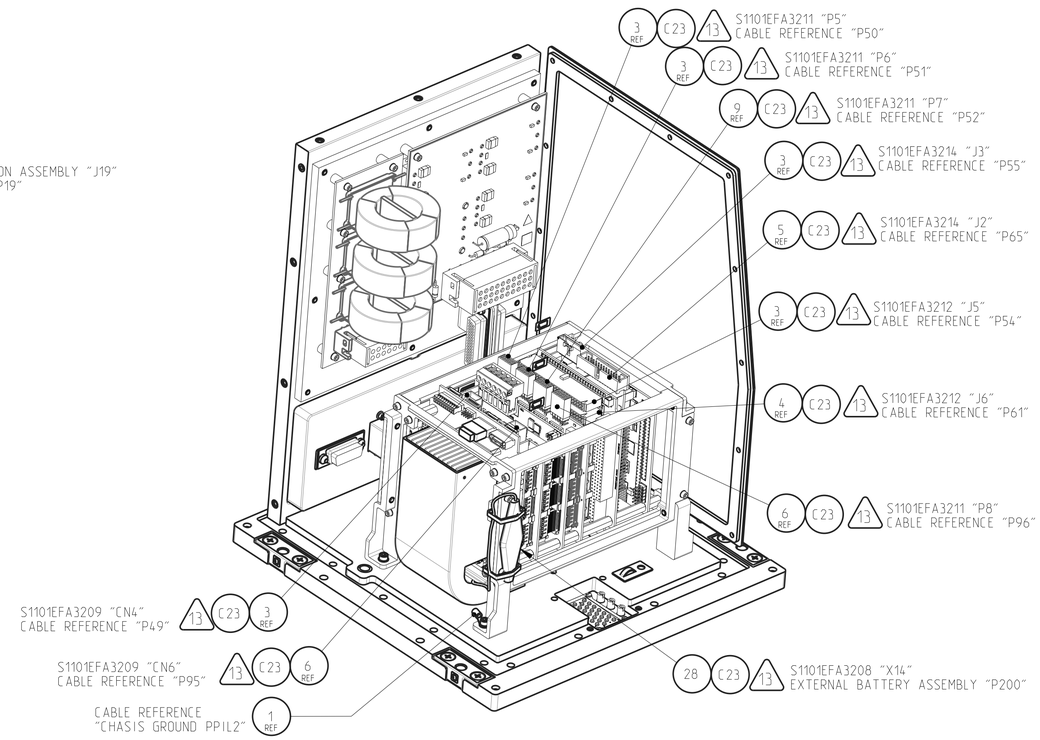
NOTES:

1. SEE SHEET 1 FOR ADDITIONAL INFORMATION. GENERAL NOTES AND TOLERANCES.



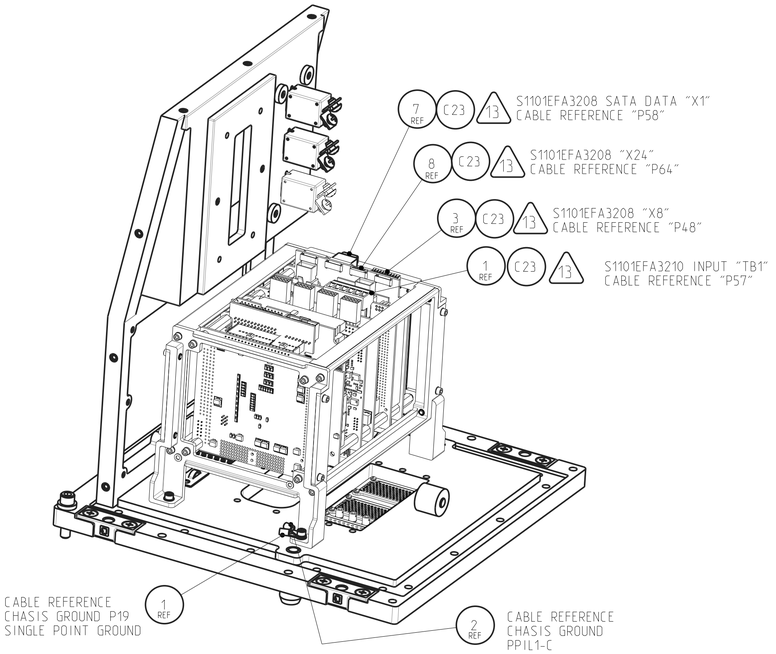
REAR INTERFACE COMPONENTS VIEW

NOTE: SOME ITEMS REMOVED FOR CLARITY



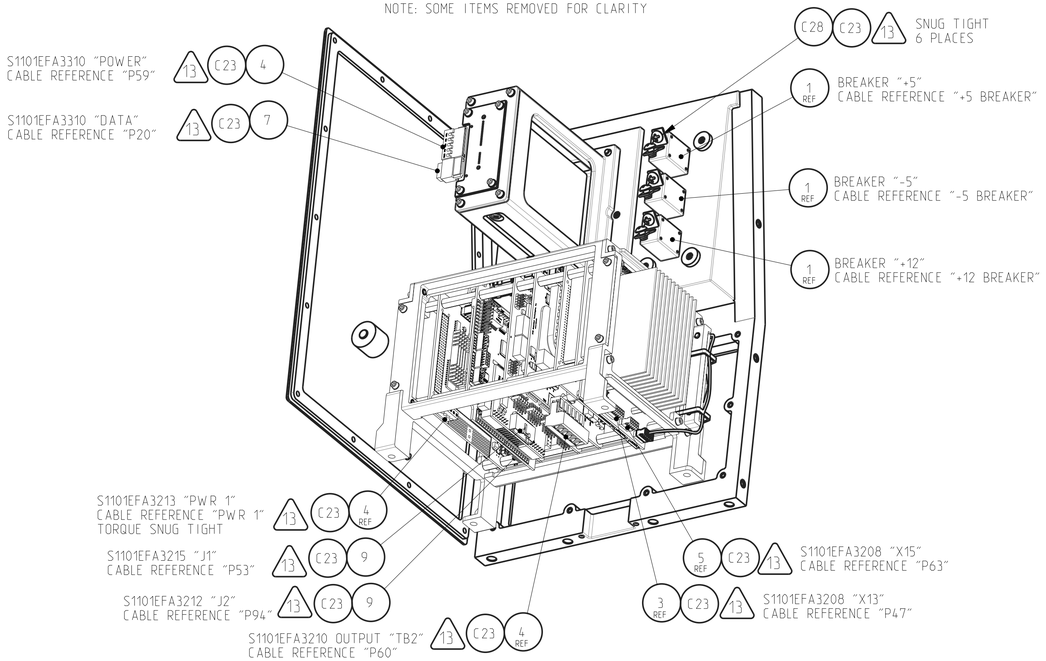
HARNESS COMPONENTS ASSEMBLY
FRONT VIEW

NOTE: SOME ITEMS REMOVED FOR CLARITY



HARNESS COMPONENTS ASSEMBLY
REAR VIEW

NOTE: SOME ITEMS REMOVED FOR CLARITY



HARNESS COMPONENTS ASSEMBLY
BOTTOM VIEW

NOTE: SOME ITEMS REMOVED FOR CLARITY

ACME-ECO-682 In Process

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA3000	C
SCALE 1/2	INITIAL REL DATE 4/29/13	SHEET 5 OF 6	

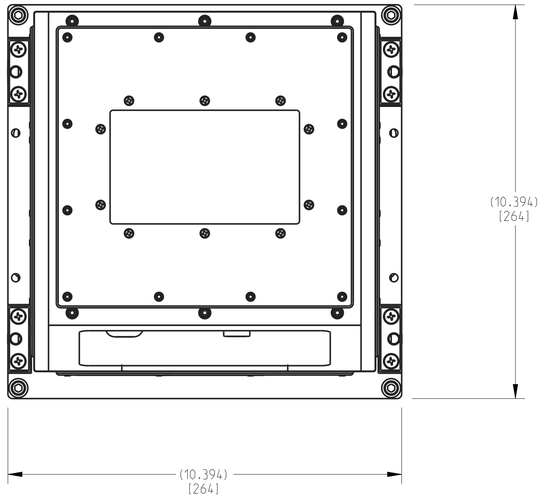
PRO/E VERSION: Wildfire 2.0

DWG NO S1101MFA3000 | SH 5 | REV C

ZIN_D_2012.frm

NOTES:

1. SEE SHEET 1 FOR ADDITIONAL INFORMATION. GENERAL NOTES AND TOLERANCES.



(NOTE: BREAKER COVER ASSEMBLY NOT SHOWN)

ACME-ECO-682 In Process

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA3000	C
SCALE 1/2	INITIAL REL DATE 4/29/13	SHEET 6 OF 6	

PRO/E VERSION: Wildfire 2.0

DWG NO S1101MFA3000 SH 6 REV C

ZIN_D_2012.frm